Released-Materials Reproduction, Study 1 (`or1-`)

Running the Anthropic Jacobian-lens battery on Qwen 3.6 27B and OLMo 3.1 32B Instruct

development/methods tier — prompt-exact where released, method-reconstructed where specified, gated where not identified

J-space campaign — official-repro sideline

branch `interp_jspace_official_repro_1` — upstream `anthropics/jacobian-lens @ 581d3986`

2026-08-08 (OR1-A Qwen block)

Abstract

We run the released Jacobian-lens evaluations (6 sets) and experiment prompt sets (11 sets) from the public companion release of "Verbalizable Representations Form a Global Workspace in Language Models" under one audited harness on two open-weight models, and separate prompt, lens, intervention, capability, and model effects. This report covers the Qwen lane: the campaign-pinned checkpoint with the published n=1000 lens — the study's mechanical instrument anchor (OR-Q1) and its first generalization surface (OR-Q2). Fidelity classes and gate states are carried on every cell; gated cells are missing, never zero; no confirmatory language is licensed anywhere in Study 1.

1 Provenance and instrument identity

Upstream release	`anthropics/jacobian-lens` @ `581d3986` (11 experiments + 6 evals, hash-pinned)
Qwen checkpoint	`Qwen/Qwen3.6-27B` @ `6a9e13bd` (64 layers, d=5120, 48 GDN blocks verified)
Published lens	`neuronpedia/jacobian-lens` @ `a4114d7`, n=1000, sha256 `1718c8c5...` byte-verified
Readout parity	max abs. diff 0.0e+00 (hard stop threshold 10^{-4}; exact top-20 order agreement)
α=0 no-op	exact (0.0 logit difference across a full-position band session)
g-folding audit	min cosine 0.382 over 26 battery tokens × 14 band layers — **material**; D9: paper-literal unfolded v_t stays prima
Bands	paper grid 25 layers; workspace band $[24, 58]$; campaign band sensitivity $[20 \ldots 44]$

Table 1: Lane admission. The boot-probe sentinel (Fig. 1) reads the latent country mid-band and the currency near the output — the instrument behaves as the paper describes before any experiment runs.

2 The paper's own example, on Qwen

3 Six released lens evaluations (OR-Q1)

4 Qwen causal core (OR-Q2)

4.1 Verbal report

Incident note (or1-001): the v1 numbers below score the leading-space token form at a boundary whose in-context form is bare; they *understate* the effect (e.g. the `color:Red` swap made the model's top-1 literally `Red` while the scored `␣Red` form sat at rank 5). A v2 rerun with min-over-forms scoring supersedes these at the next Qwen GPU slot; see `reports/INCIDENT_or1-001_vr_token_form.md`.

Release-literal rule (first ten listed candidates, skip the answer), swap-out = greedy token at the generation boundary (D1), α=1 across the band at every prompt position. Category-equal top-1 0.0% and top-5 0.8% over 14 categories (118 executed trials; 64 trials improved the candidate's rank, 53 worsened). Paper context

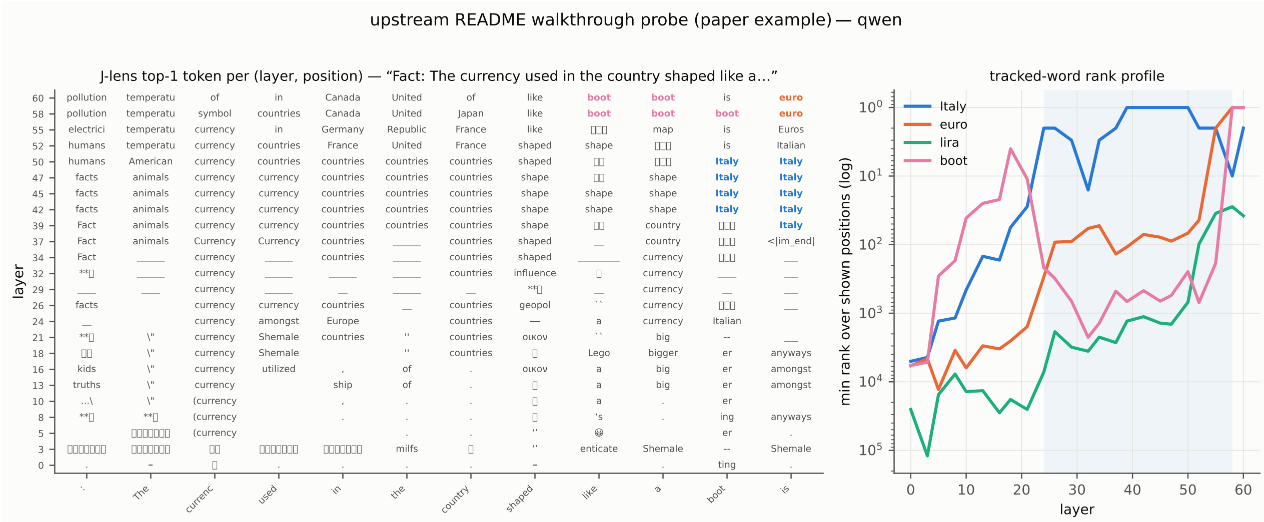

Figure 1: The upstream README walkthrough probe ("The currency used in the country shaped like a boot is") under the published lens on Qwen 3.6 27B. Left: J-lens top-1 token per (layer, position); colored cells are tracked words. Right: tracked-word best rank by layer (shaded: workspace band). Italy appears as latent content mid-band and hands off to euro/lira toward the output — the paper's slice-visualisation phenomenology, reproduced from released materials.

	Jacobian lens			logit lens		
Set	@1	@5	@20	@1	@5	@20
multihop (n=93)	0.244	0.399	0.641	0.173	0.327	0.542
multilingual (n=107)	0.237	0.384	0.580	0.184	0.327	0.497
poetry (n=98)	0.010	0.020	0.071	0.020	0.031	0.061
typo (n=96)	0.135	0.271	0.500	0.333	0.594	0.719
order-ops (n=55)	0.145	0.455	0.727	0.209	0.600	0.727
association (n=102)	0.039	0.059	0.108	0.010	0.059	0.118

Table 2: pass@k, token-valid primary. All-source-layer sensitivity and released-denominator sensitivity live in the registered JSON.

(Claude): 88% top-5. The effect on Qwen under the paper-literal operation is *directionally present in rank movement but far below the paper's report-rate*: Fig. 5.

4.2 Flexible generalization

All 192 ordered off-diagonal swaps; capability-conditioned primary (source and target baselines correct; baseline accuracy 51.6%); α=2 full sensitivity. Category-equal capable top-1: 5.8% (α=1) → 0.0% (α=2); diagnostic all-executable: 5.6%. Paper context: 76/192 → 101/192 (Sonnet 4.5). Per-category grids in Fig. 6.

4.3 Probe-swap, raw J-lens token arm

Fidelity `prompt_exact_representation_adapted_raw_jlens` (the paper's own §3.3 headline arm; the official linear-probe arm is NOT-IDENTIFIED — its training corpus is absent from the release). Population: 90 released items, 7 tokenization-gated, 83 executed, 63 baseline-capable. Capable top-1 14.3% (diagnostic 14.5%); multihop 10.5% vs non-multihop 15.9%. Paper context: 60% (Claude, n=90).

5 Worked examples: what the swaps actually do

The aggregate rates hide a bimodal reality worth seeing item by item. When the paper-literal swap works on Qwen, it works emphatically — the predicted counterfactual answer jumps hundreds of ranks to top-1:

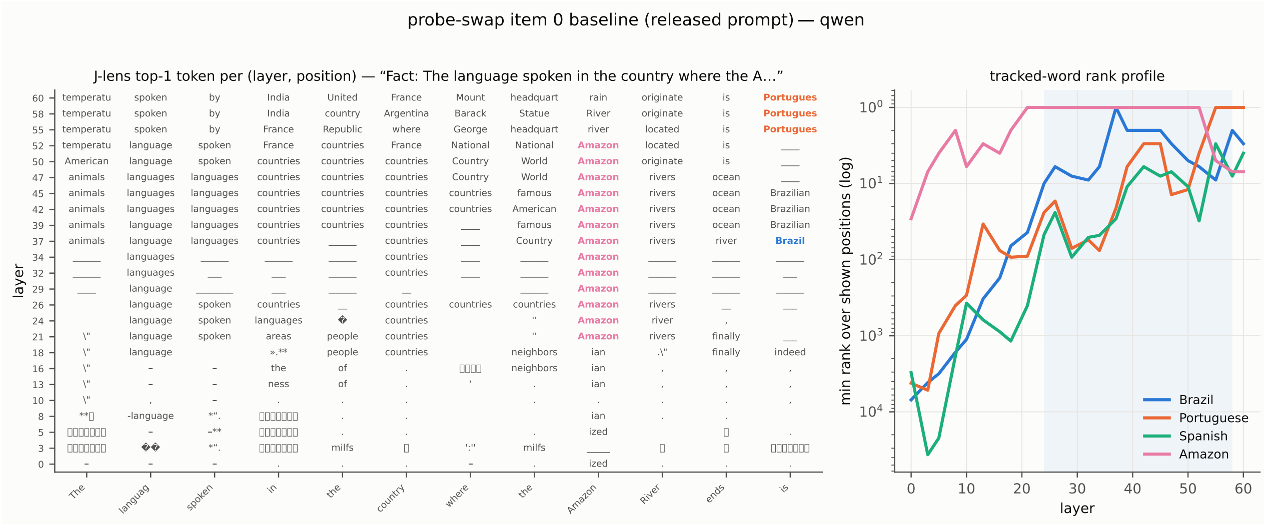

Figure 2: Released probe-swap item 0 (“the language spoken in the country where the Amazon River ends”) — the two-hop structure is visible: Brazil (bridge entity) carried in-band while the output side resolves toward Portuguese.

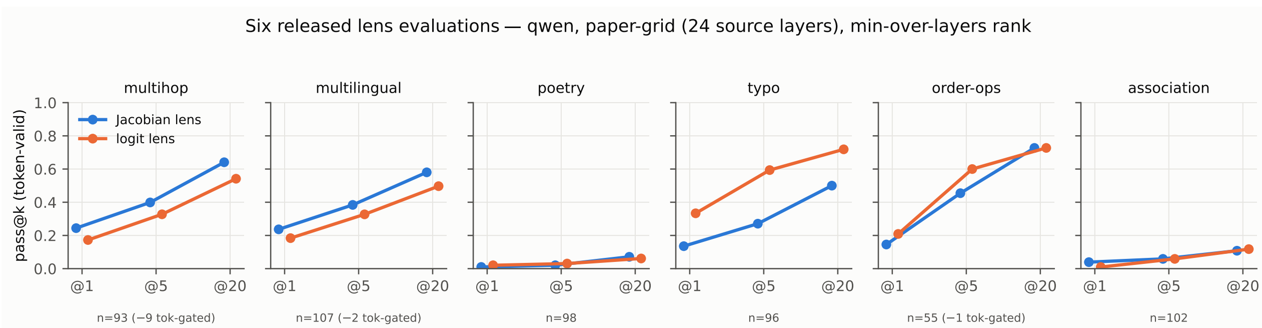

Figure 3: pass@k per released set, token-valid denominators, min-over-24-paper-grid-layers at the released readout position.

Two structural observations. First, successes cluster where the swapped-in argument has a strong, single-token, high-frequency identity (cities, countries, elements); the sparse-identity categories (months as time-words, small numbers) never flip. Second, the near-miss mass (top-5 but not top-1) plus the incident-or1-001 form effect mean the honest Qwen picture is “the mechanism visibly engages but rarely completes the flip at α=1” — which makes the α=2 *reversal* (0% capable) the more interesting anomaly: on Qwen the doubled patch overshoots into off-distribution territory rather than completing near-misses, unlike the paper’s Sonnet where α=2 lifted 76/192 to 101/192.

6 Interpretation under the wording router

Conformance and admission passed mechanically (exact parity, exact no-ops, verified pins), so every result above is a scientific observation about this model/instrument cell, not a harness state. The lens-eval layer profiles reproduce the paper’s methods-comparison direction on the latent-content distributions; the paper-literal α=1 coordinate-swap core is *directionally present but substantially weaker on Qwen 3.6 27B than the paper’s Claude values*, with rank movement concentrated below top-1. Maximum licensed sentence (plan §7.6 routing, pre-OLMo): *“conformance passes; the Qwen core is a valid generalization result”* — the released functional properties do not transfer at paper magnitude to this cell under prompt-exact, method-reconstructed conditions. No workspace-existence or -absence claim is licensed; OLMo, the extended battery, and the instrument cross-over (which will separate intervention semantics from prompt population) follow in OR1-

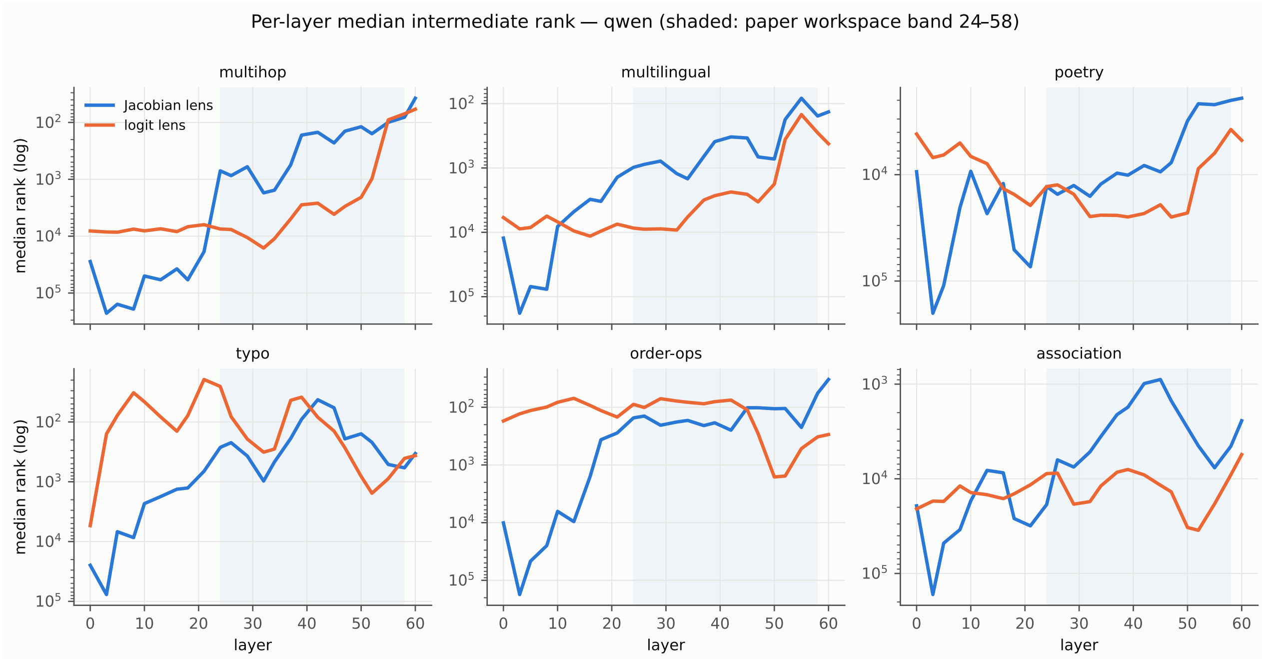

Figure 4: Median intermediate rank by layer. On the latent-content sets (multihop, multilingual) the J-lens reaches useful ranks by the band onset while the logit lens remains near rank 10^4 until ∼L45 — the paper's central methods-comparison claim, visible on an open model. On typo the logit lens is stronger (the correction is surface-adjacent); order-ops numbers sit in the prompt, so the logit lens copies them early.

B/C/D.

7 Population accounting

Every released row carries source-present / tokenization-valid / geometry-valid / baseline-capable / executed states in the registered raw records (`*_raw.jsonl`, plan §12 schema); the attrition waterfall figure ships with the OLMo lane so both lanes share one accounting frame. Nothing gated was encoded as zero anywhere in this report.

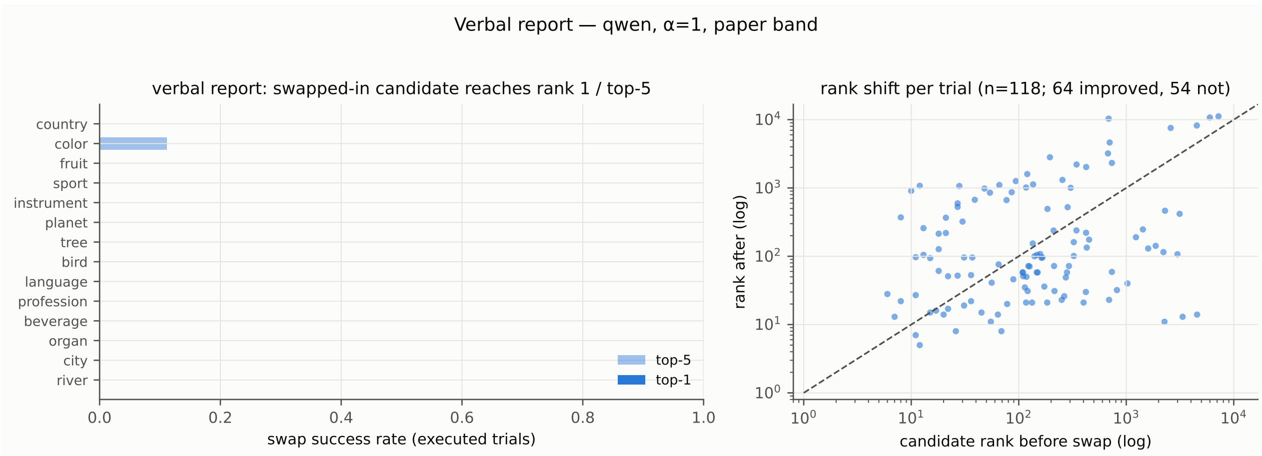

Figure 5: Verbal report on Qwen: per-category success rates (left) and per-trial rank shifts (right; below the diagonal = swap moved the candidate toward the output).

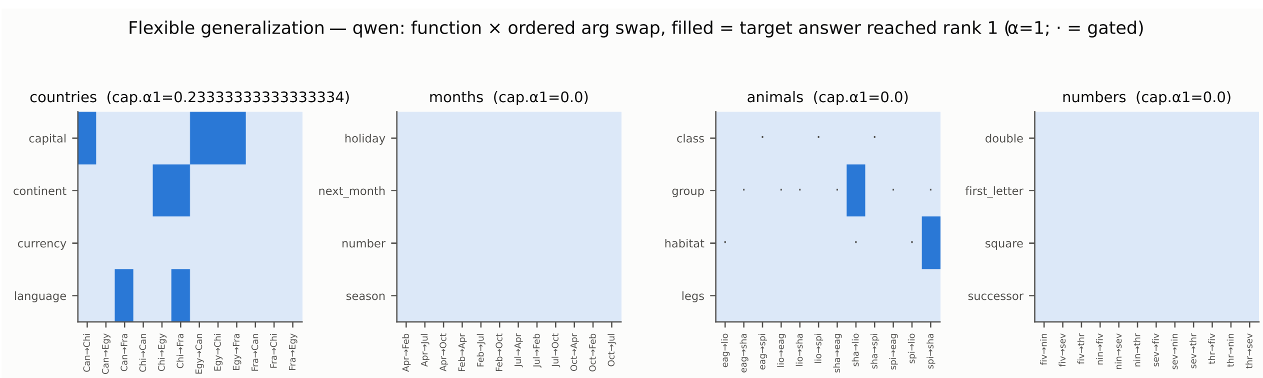

Figure 6: Function × ordered-argument-swap outcomes, α=1 (filled = target answer reached rank 1).

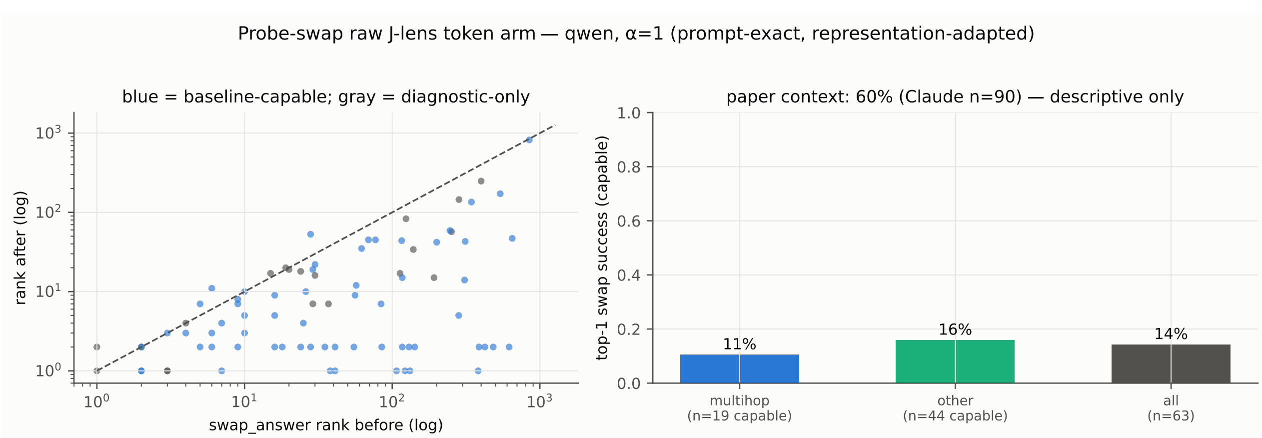

Figure 7: Probe-swap raw arm on Qwen: per-item rank movement of the predicted swapped answer (left) and capable top-1 by category group (right).

item	category	swap-answer rank before	after	model output after swap
`ex-element-symbol-11-26`	element-symbol	382	1	`:`
`ex2-city-continent-Sydney`	city-continent	132	1	`South`
`ex2-city-capital-Munich`	city-capital	122	1	`Tokyo`
`ex2-city-language-Moscow`	city-language	107	1	`Russian`
`ex-city-capital-Lyon-Naples`	city-capital	41	1	`Paris`
`ex-city-language-Lyon-Naples`	city-language	38	1	`French`
`amazon-language`	multihop	7	1	`Spanish`
`ex-element-state-8-26`	element-state	3	1	`solid`

Table 3: The 12 probe-swap successes (of 63 baseline-capable items), largest baseline distance first. `ex2-city-capital-Munich`: swapping the J-lens vector Munich→(released target) at every prompt position moves the predicted capital from rank 122 to the model's literal output. One success (`rhyme-rain-neighbor`) was already rank 1 before the swap (1 such trivial cell in the count); 31 further items land in the top-5 without reaching top-1 — the modal failure is *under-strength*, not wrong-direction.

category	function	swap	rank before	output
countries	capital	Egypt→Canada	480	`Ottawa`
countries	capital	Canada→China	338	`Beijing`
countries	capital	Egypt→China	297	`Beijing`
animals	group	shark→lion	275	`pride`
countries	language	China→France	203	`French`
animals	habitat	spider→shark	103	`ocean`
countries	capital	Egypt→France	47	`Paris`
countries	continent	China→Egypt	14	`Africa`

Table 4: The 10 flexible-generalization α=1 successes concentrate in the *countries* category (capable top-1 23.3% there): `capital Egypt→Canada` drives `Ottawa` from rank 480 to the output. Months and numbers show none — the four released categories differ enormously, exactly the per-category variance the paper's own appendix warns about.